HRCSR: Hierarchical Risk-Controlled Source Routing

Global source evidence -> source-specific knowledge -> event routing -> out-of-sample abstention

Candidate source pool

$D_1, \ldots, D_K$
heterogeneous temporal graphs

Level 1: RASP portfolio

risk-adjusted target-train utility
selects $\mathcal{S}^*$ and source priors π_s

Level 2: Expert bank

one residual expert per selected source
$\Delta_s(h_t) = E_s(h_t)$; no target expert

Level 3: Query router

shared response-aware scorer $\alpha_{x,t,s}$
local entropy-aware acceptance $a_{x,t}$

Level 4: OOS LCB gate

bootstrap lower bound L_T
$G_T = \mathbb{1}[L_T > 0]$

Safe target representation

$$h'_{x,t} = h_{x,t} + I_{x,t} G_T a_{x,t} \tanh(\rho_R) \sum_s \alpha_{x,t,s} \Delta_s$$

Target-training chronology used by the safety decision

Optimization prefix (gradient updates)

fit target adapter, link head, and router
path-neutral route + identity objectives
per-event pairwise regret discourages harmful routing

Held-out calibration suffix (no gradients)

historical collision-free negatives
$g_i = RR_i^{route} - RR_i^{identity}$
chronological windows + bootstrap LCB

Validation / test

not used in gate statistic
validation may select epoch

Decision semantics: from whom? -> what knowledge? -> which source for this query? -> should specialized transfer be trusted?